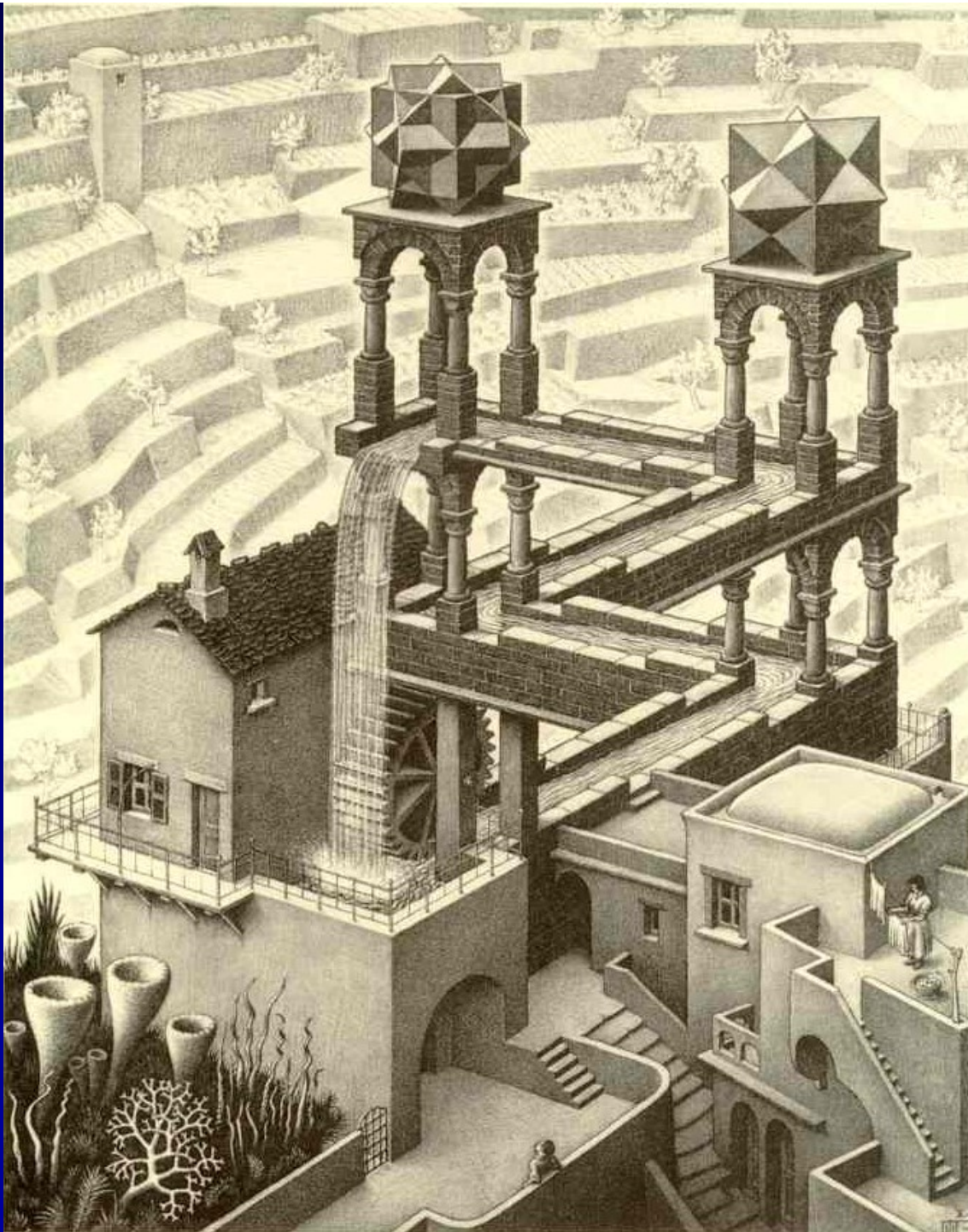




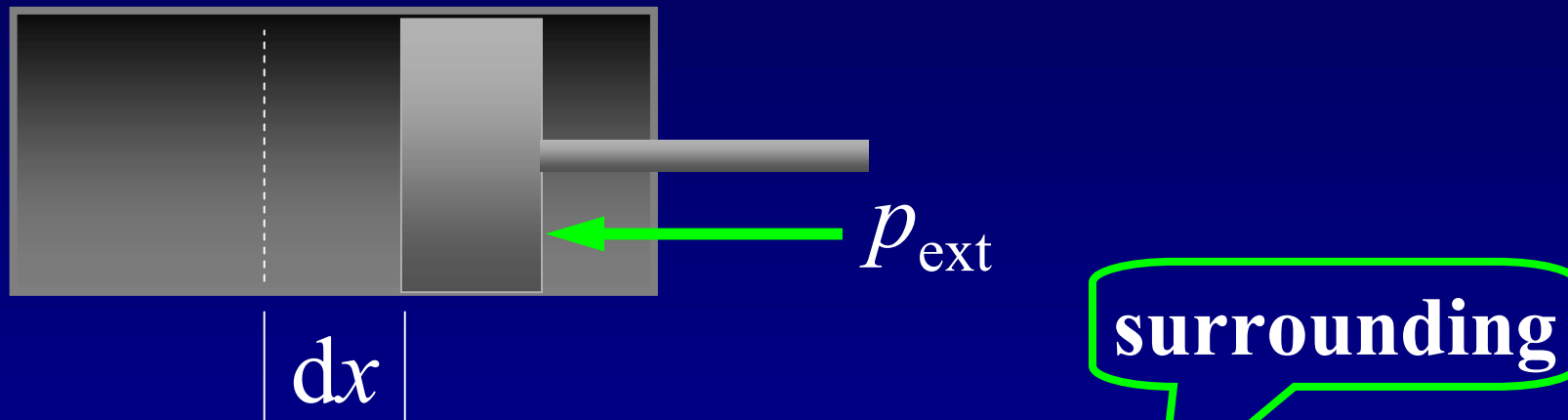
Escher visual illusions

Chapter 11

The First Law of Thermodynamics

11.1 Work and internal energy

work done by the surrounding to the system

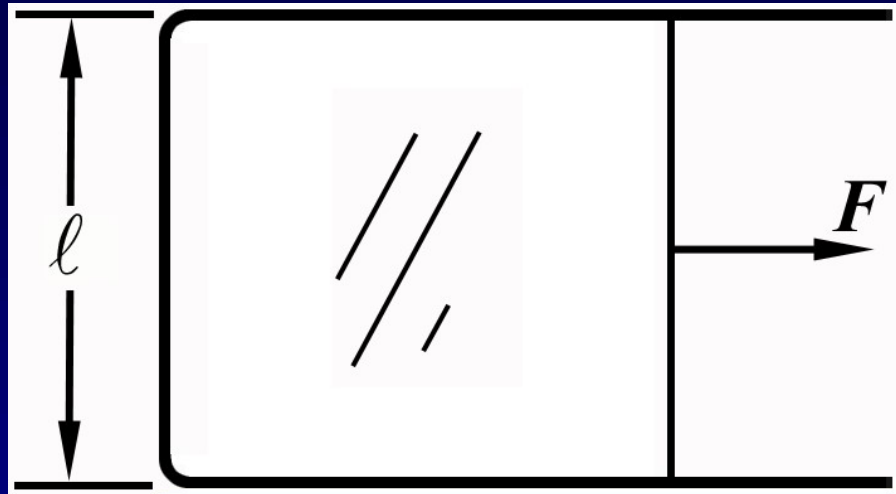


$$dW = p_{\text{ext}} A dx = p_{\text{ext}} dV_S = -p_{\text{ext}} dV$$

In quasi-static process $p = p_{\text{ext}}$ so

$$dW = -p dV$$

For a surface system of tension σ



$$\begin{aligned}dW &= Fdx \\ &= 2\sigma \ell dx \\ &= \sigma dA\end{aligned}$$

generalized displacement

$$dW = Y_1 dX_1 + Y_2 dX_2 + \dots$$

generalized force

external parameter

$Y: -p, J, \sigma, E, H, \varepsilon, \mu$

$X: V, L, A, P, M, Q, n$

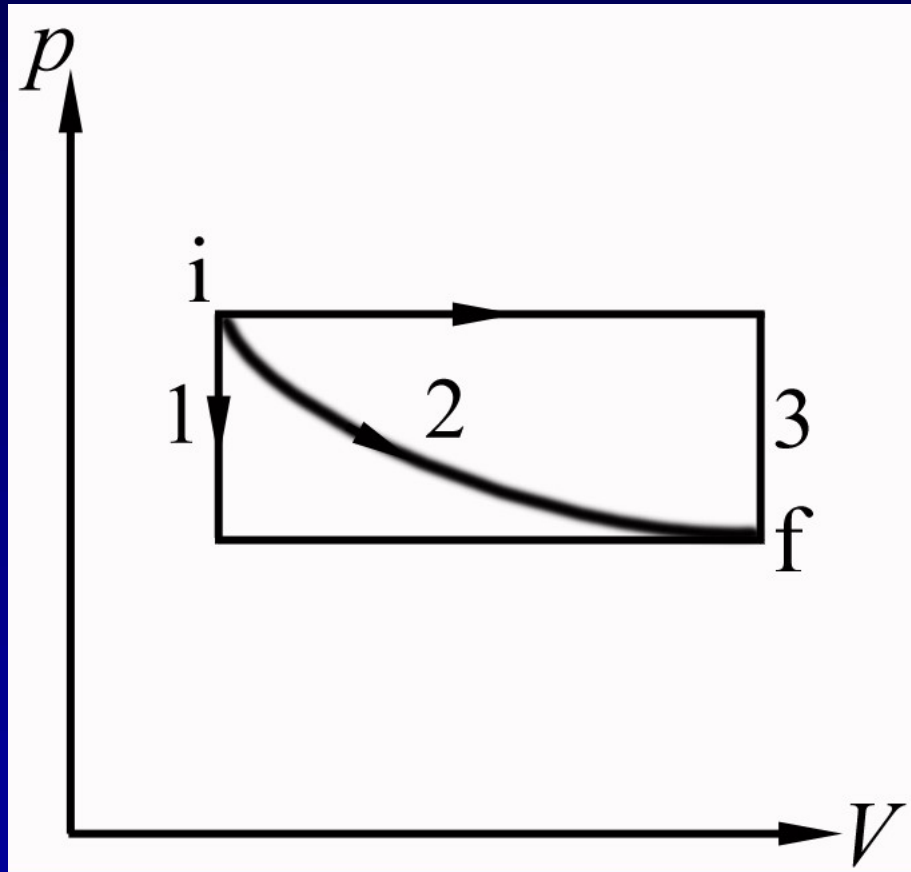


electrostriction and piezo effect

magnetostriction and piezomagnetic effect

$(-p, V)$ as representative

Work is **path**-dependent



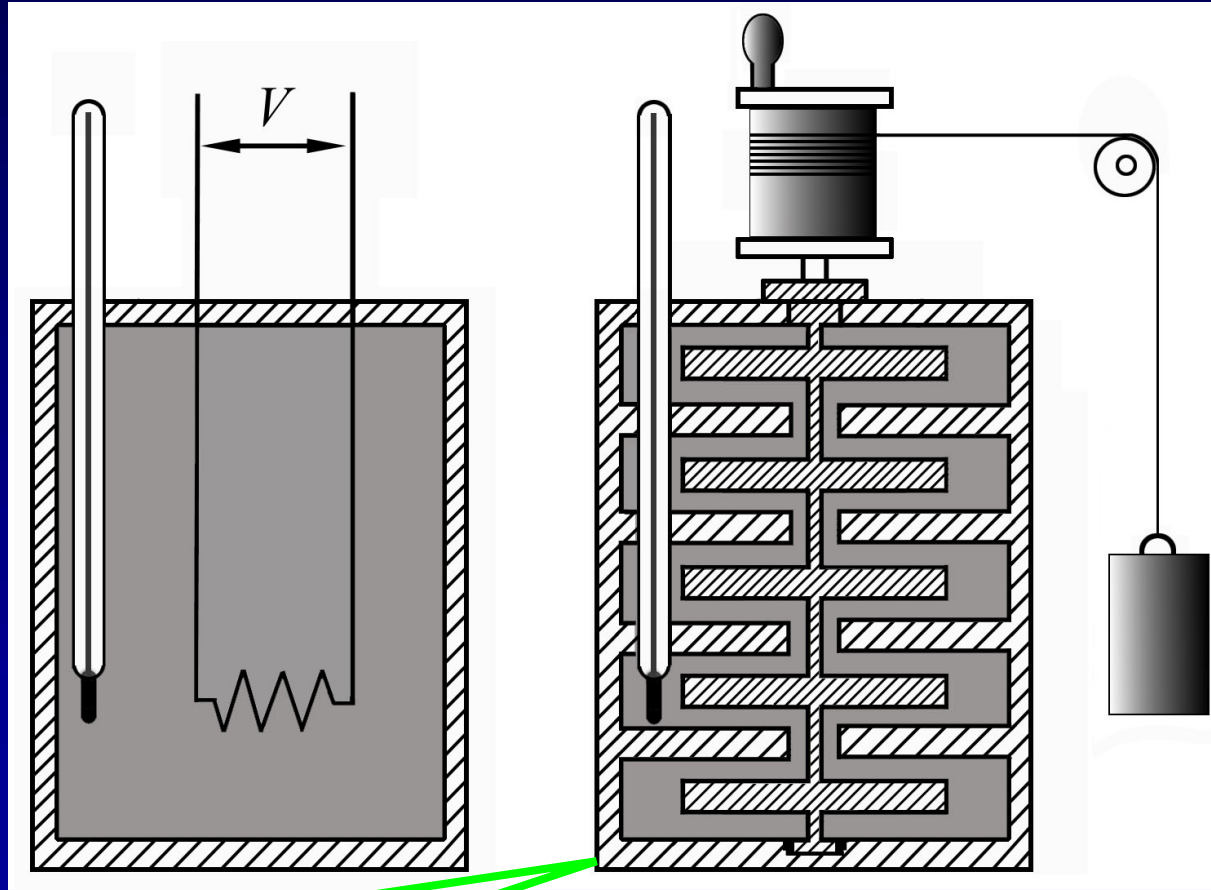
$$W = - \int_{V_i}^{V_f} p dV$$

$$W_1 = -p_f(V_f - V_i)$$

$$W_3 = -p_i(V_f - V_i)$$

$$W_2 \sim -\ln \frac{V_f}{V_i}$$

In mechanics, work done by conservative force is **path-independent**. In Joule's experiments,



adiabatic envelope

mechanical, electric, magnetic...
quasi-static or not

Work done is **same** for all adiabatic paths
connecting an initial and a final state.

Internal energy function is defined through

$$\Delta U = U_B - U_A \equiv W_{BA}$$

If $A \xrightarrow{\quad} B$,

$$U_B - U_A \equiv -W_{AB}$$

Never write $U_B + W_S = U_A$

11.2 Heat and the first law of thermodynamics

Caloric theory: caloric fluid

热质

- is massless
- is colorless
- conserved in its total amount in the universe
- flows from hotter to colder bodies and
- can not be separated from objects

Mechanism of steam engine?

heat capacity, latent heat — misleading terms

‘problem phenomena’

Benjamin Thompson(later Count Rumford of Bavaria) asserted that **heat is related to motion, work. (1798)**

Joule’s mechanical equivalent of heat

$$1 \text{ cal}_{15} = 4.1858\text{J}$$

In general

$$\Delta U \neq W \quad \text{unless} \quad W = W_s$$

$Q \equiv U_f - U_i - W$ is defined as **heat** released by surrounding to system

$$\Delta U = W + Q$$

$$dU = \delta Q + \delta W$$

the first law of thermodynamics

Another statement of the first law is:

• *The perpetual motion machines of the first kind are impossible*

cycle $\Delta U = 0, \quad Q = 0, \quad W \neq 0$

The first law is really the law of energy conservation.

German physician J R **Mayer** (1814–1878) In 1842, a private pamphlet, *Organic Motion in Its Connection with Nutrition* listed 25 forms of energy.

J P **Joule** (1818–1889) In 1840, he sent a paper entitled 'On the Production of Heat by Voltaic Electricity' to the Royal Society (22 years old).

German physician H von **Helmholtz** (1821–1894) *On the conservation of Force* (1847)

French military engineer S **Carnot** (1796–1832) After the publication of his *Reflections on the Motive Power of Fire* in **1824**, it became clear that Carnot had arrived at the conservation of energy before all the others.

11.3 Heat capacity and specific heat

$$C = \lim_{\Delta T \rightarrow 0} \frac{\Delta Q}{\Delta T}$$

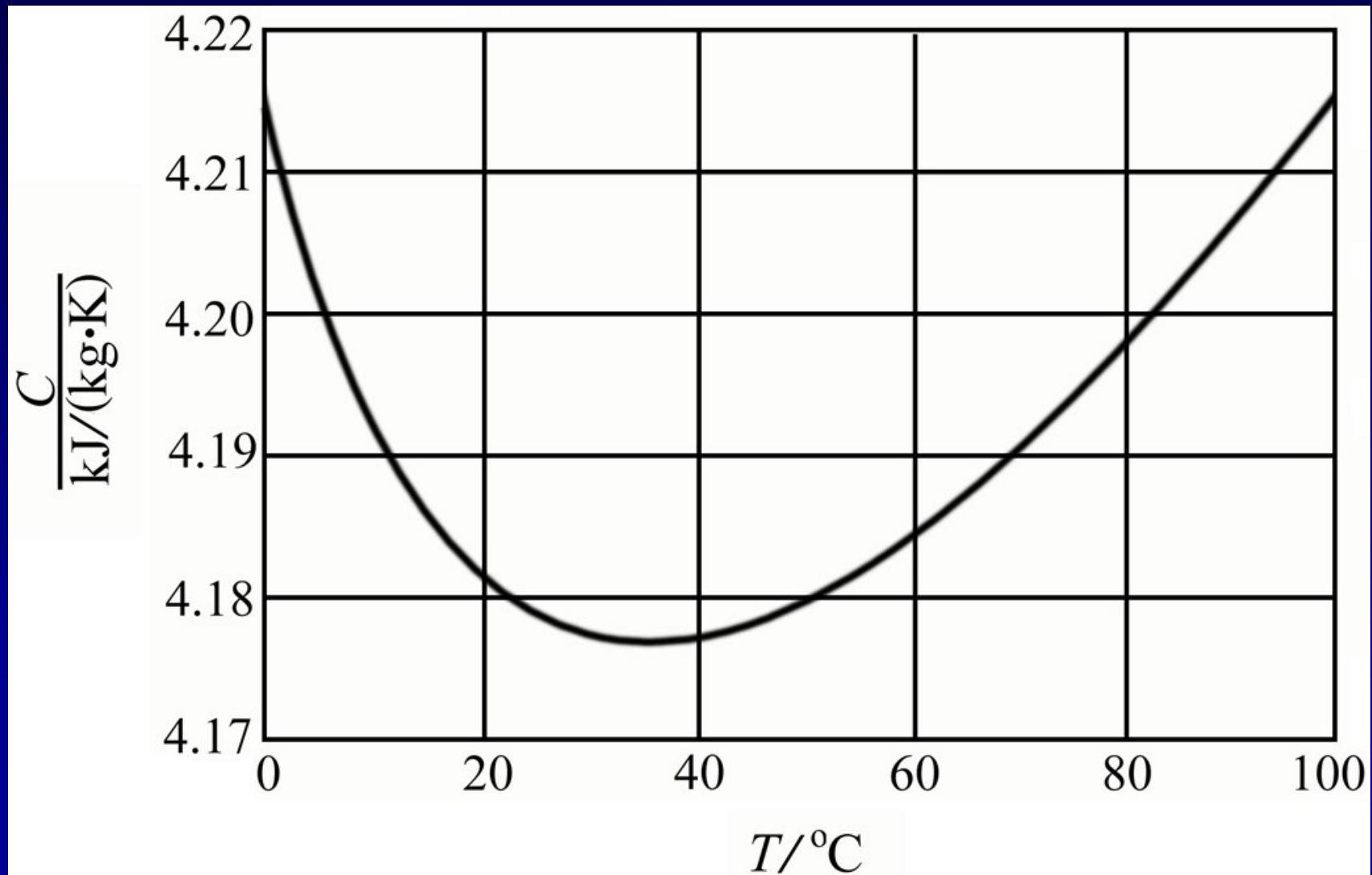
C/m **specific heat** capacity

$C_m \equiv C/n$ **molar heat** capacity

Heat capacity is process-dependent

- $C > 0$ (absorption and $\Delta T > 0$);
- $C < 0$ (emission and $\Delta T > 0$, gravitational system such as black hole),
- $C = 0$ (adiabatic);
- $C = \infty$ (isothermal, heat reservoir)

Heat capacity is temperature-dependent.



$$C_V = \lim_{\Delta T \rightarrow 0} \left(\frac{\Delta Q}{\Delta T} \right)_V = \lim_{\Delta T \rightarrow 0} \left(\frac{\Delta U}{\Delta T} \right)_V = \left(\frac{\partial U}{\partial T} \right)_V$$

$$C_p = \lim_{\Delta T \rightarrow 0} \left(\frac{\Delta Q}{\Delta T} \right)_p$$

$$= \lim_{\Delta T \rightarrow 0} \left(\frac{\Delta U + p\Delta V}{\Delta T} \right)_p$$

$$= \left\{ \begin{array}{l} \left(\frac{\partial U}{\partial T} \right)_p + p \left(\frac{\partial V}{\partial T} \right)_p \\ \frac{\partial(U + pV)}{\partial T} \Big|_p \equiv \left(\frac{\partial H}{\partial T} \right)_p \end{array} \right.$$

enthalpy

焓

ratio of specific heat $\gamma = \frac{C_p}{C_V}$

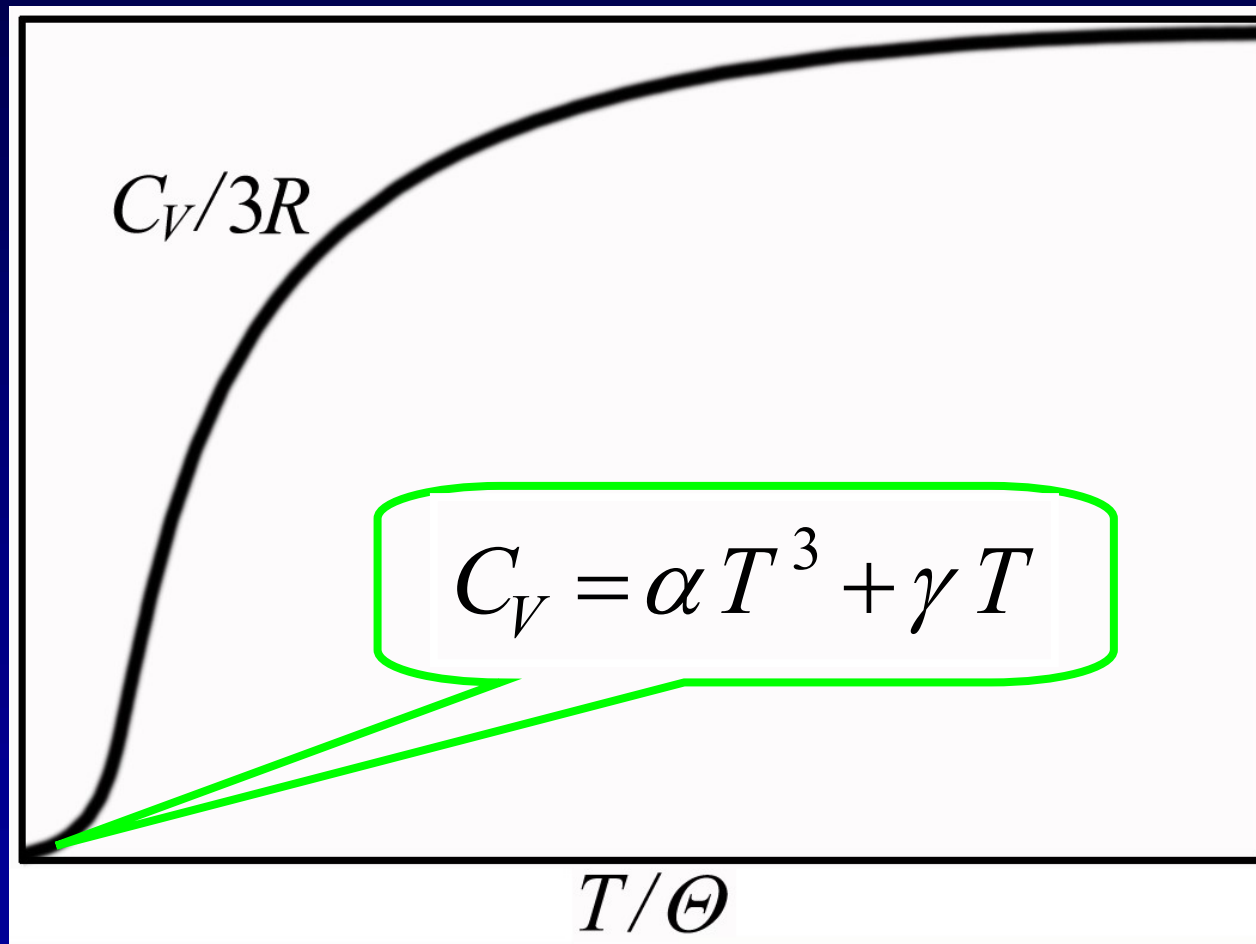
heat capacity of ideal gases

monatomic ideal gas	$C_V = \frac{3}{2}nR$
diatomic ideal gas	$C_V = \frac{5}{2}nR$
polyatomic ideal gas	$C_V = \frac{6}{2}nR$

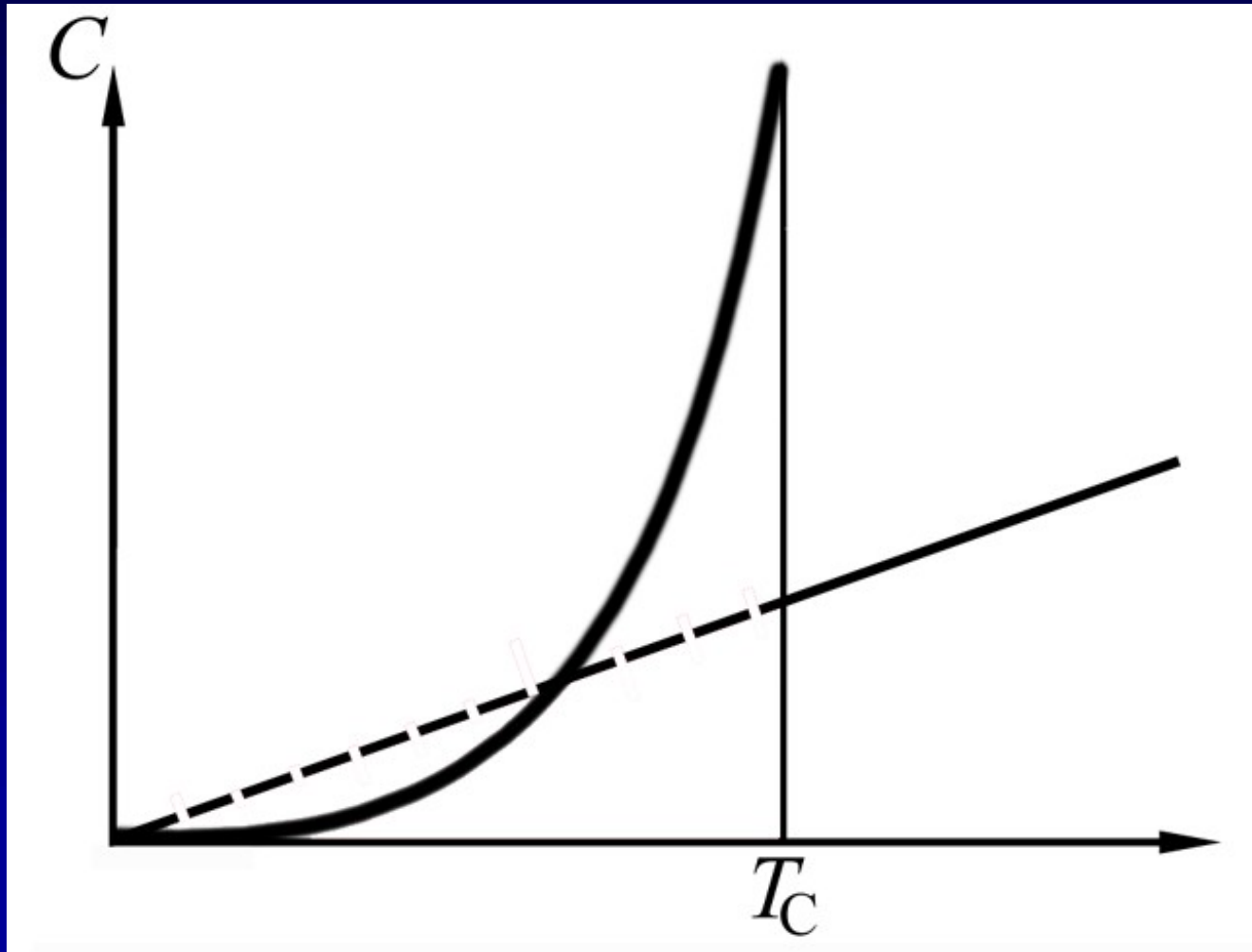
From ①...②...

Study of specific heat is a very important way to get insight of matters, especially when phase transitions happen.

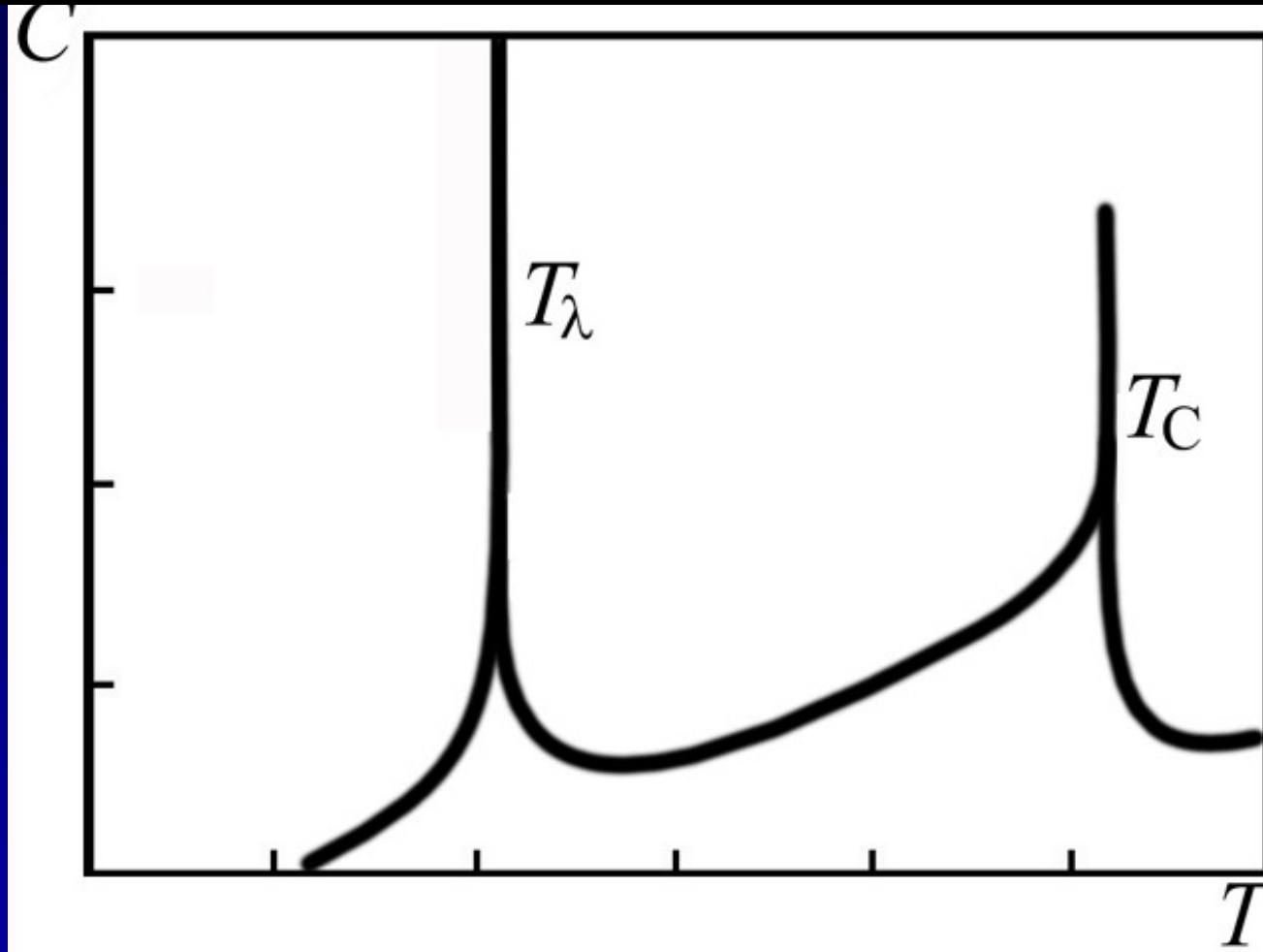
Molar heat capacity of solids



Superconducting electron specific heat



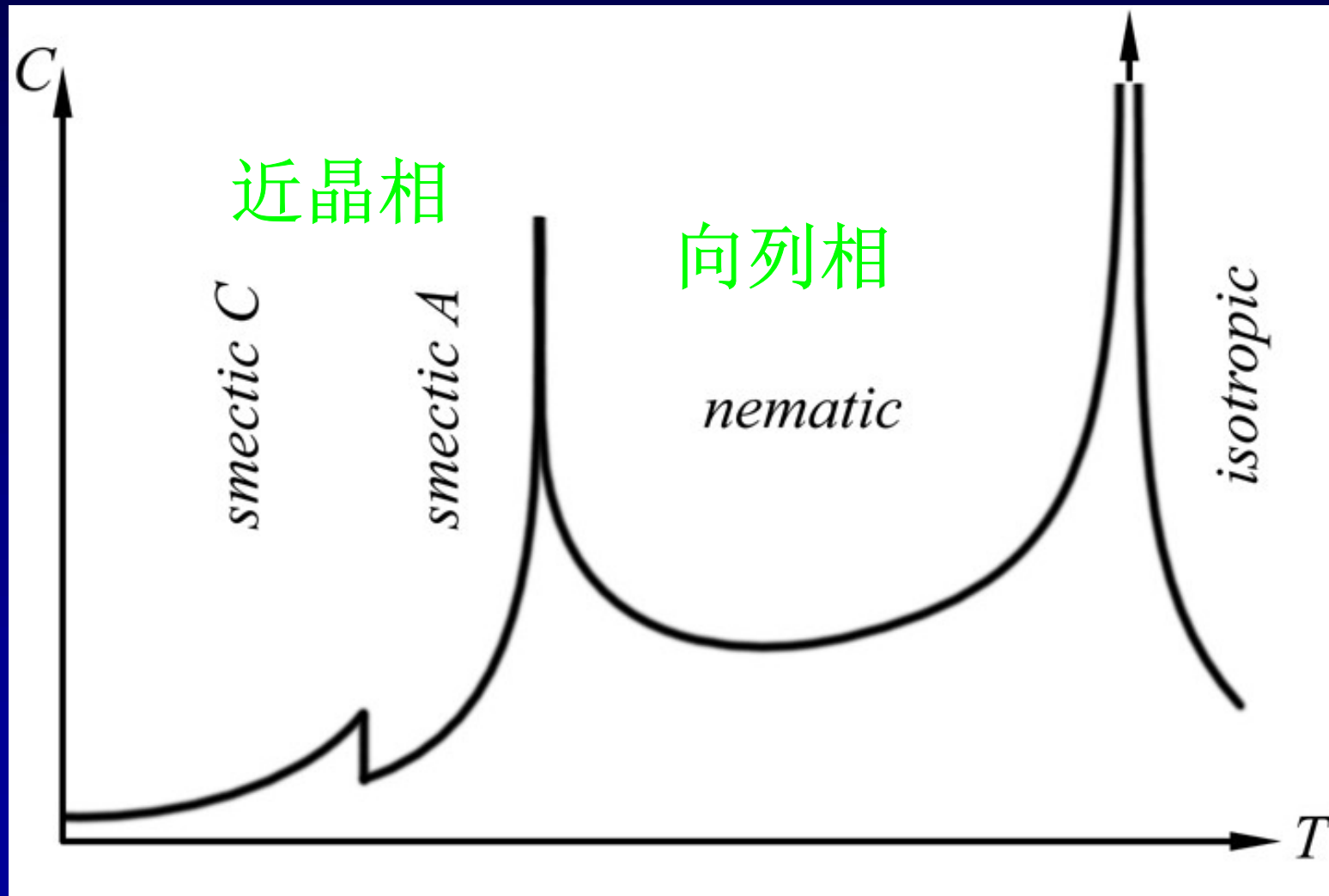
low temperature specific heat of ^4He



boiling point of 4.22 K

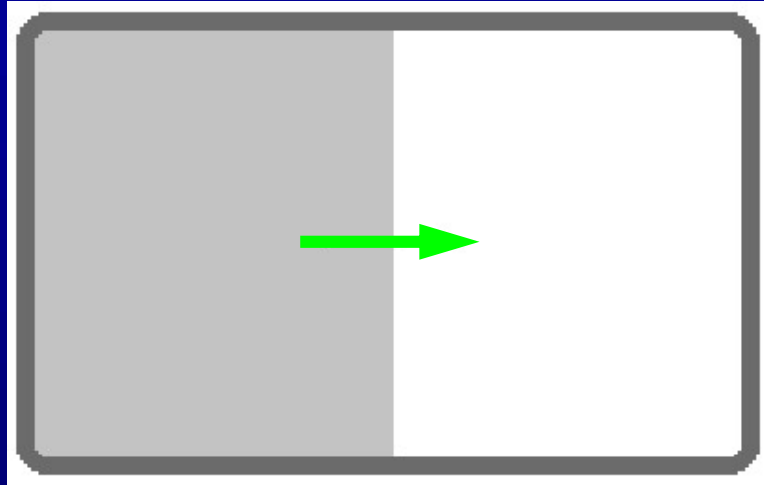
λ (lambda) point of 2.1768 K superfluid

Specific heat of liquid crystal



Assignment: 11.1, 11.2

11.4 Free expansion and internal energy of gas



$$U = U(T, V)$$

$$dU = \left(\frac{\partial U}{\partial T} \right)_V dT + \left(\frac{\partial U}{\partial V} \right)_T dV$$

$$T = T(U, V)$$

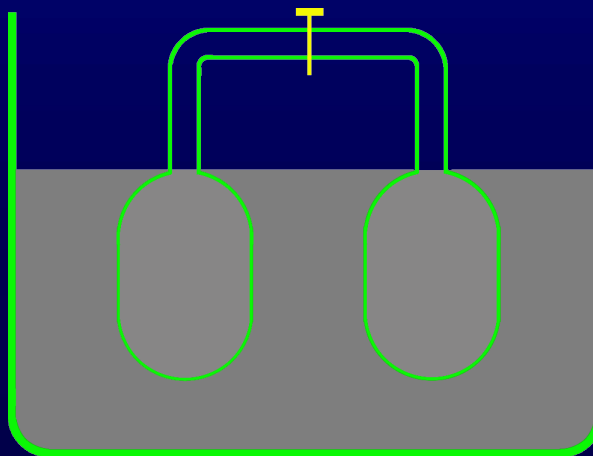
$$dU = \left(\frac{\partial U}{\partial T} \right)_V \left[\left(\frac{\partial T}{\partial U} \right)_V dU + \left(\frac{\partial T}{\partial V} \right)_U dV \right] + \left(\frac{\partial U}{\partial V} \right)_T dV$$

$$\left(\frac{\partial U}{\partial T}\right)_V = \frac{1}{\left(\frac{\partial T}{\partial U}\right)_V}$$

Joule coefficient

$$\left(\frac{\partial U}{\partial V}\right)_T = -\left(\frac{\partial U}{\partial T}\right)_V \left(\frac{\partial T}{\partial V}\right)_U = -C_V \left(\frac{\partial T}{\partial V}\right)_U$$

In 1843 Joule found $(\partial T/\partial V)_U = 0$ and deduced that for gas $U=U(T)$. (Joule's law)



Joule's law is valid only for ideal gas.

For ideal gas

$$dU(T) = \frac{dU}{dT} dT = C_V dT$$

$$U = C_V T + U_0$$

1st law $\delta Q = C_V dT + p dV$
 $= (C_V + nR) dT - V dp$

$$C_p = C_V + nR$$

$$\gamma = \frac{C_p}{C_V}$$

$$\left. \begin{array}{l} C_p = C_V + nR \\ \gamma = \frac{C_p}{C_V} \end{array} \right\}$$



$$C_V = \frac{nR}{\gamma - 1}$$

$$C_p = \frac{\gamma nR}{\gamma - 1}$$

11.5 Adiabatic equation

$$\mathrm{d}Q = 0$$

$$(\gamma - 1) \times C_V \mathrm{d}T + p \mathrm{d}V = 0$$

$$V \mathrm{d}p + p \mathrm{d}V = nR \mathrm{d}T$$

$$+) \frac{\quad}{\quad} = (\gamma - 1) C_V \mathrm{d}T$$

$$V \mathrm{d}p + \gamma p \mathrm{d}V = 0$$

$$\frac{\mathrm{d}p}{p} + \gamma \frac{\mathrm{d}V}{V} = 0$$

$$pV^\gamma = C$$

$$TV^{\gamma-1} = C$$

the 1st law
ideal gas

ideal gas

adiabatic work

$$W_S = - \int_{V_1}^{V_2} p dV = -C \int_{V_1}^{V_2} \frac{dV}{V^\gamma} \quad C = p_1 V_1^\gamma = p_2 V_2^\gamma$$

$$= \frac{C}{\gamma - 1} \left[\frac{V_2}{V_2^\gamma} - \frac{V_1}{V_1^\gamma} \right]$$

$$W_S = \frac{1}{\gamma - 1} [p_2 V_2 - p_1 V_1]$$

$$= \frac{nR}{\gamma - 1} [T_2 - T_1]$$

$$= C_V [T_2 - T_1]$$

$$U = C_V T + U_0$$

sound speed

$$v = \sqrt{\frac{B_T}{\rho}}$$

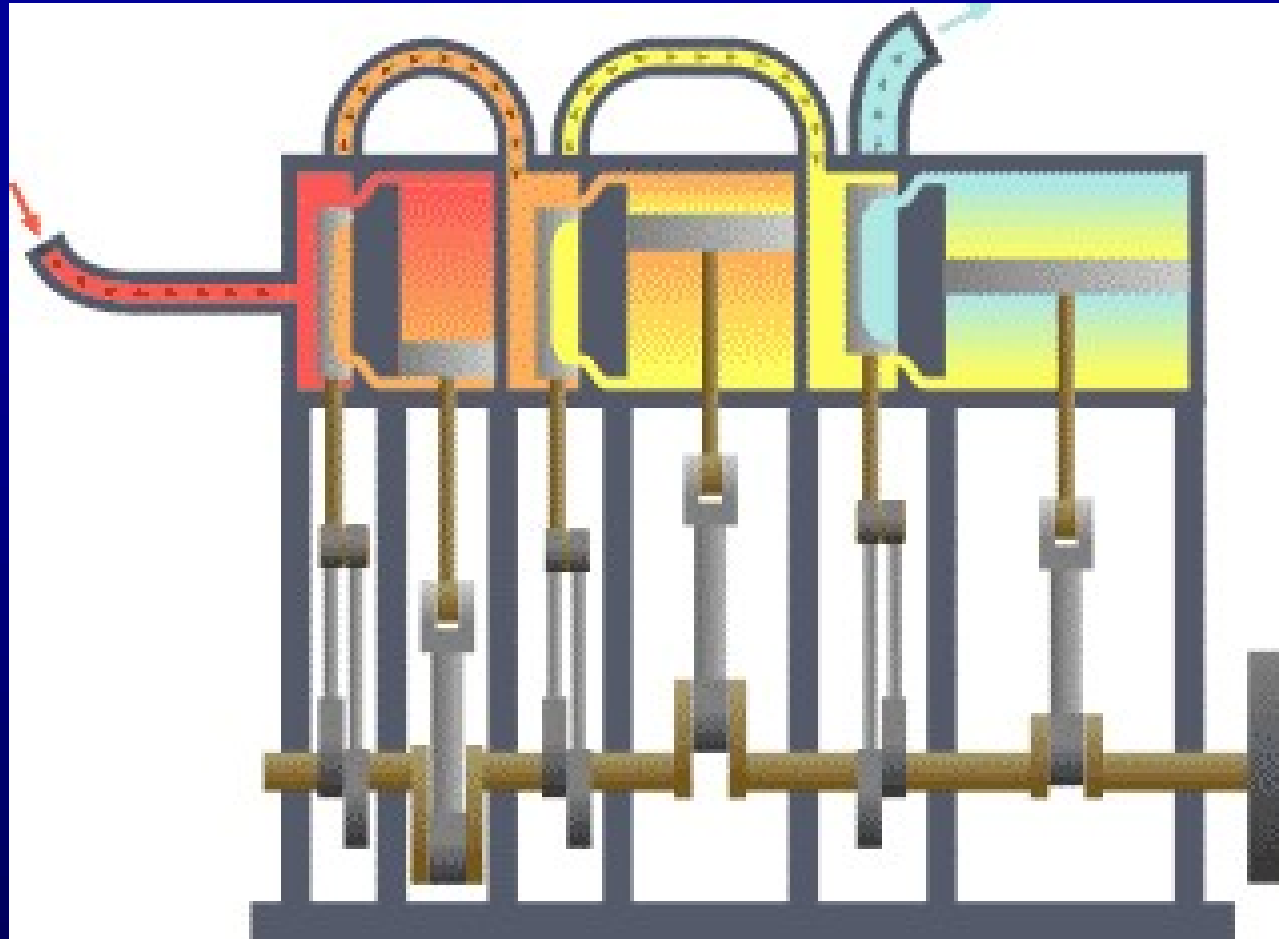
$$B_T = -V \left(\frac{\partial p}{\partial V} \right)_T = p \longrightarrow v = \sqrt{\frac{p}{\rho}}$$

(1719 Newton)

$$B_S = -V \left(\frac{\partial p}{\partial V} \right)_S = \gamma p; \longrightarrow v = \sqrt{\frac{\gamma p}{\rho}}$$

(1816 Laplace)

Heat (1798) : Rumford



cylinders

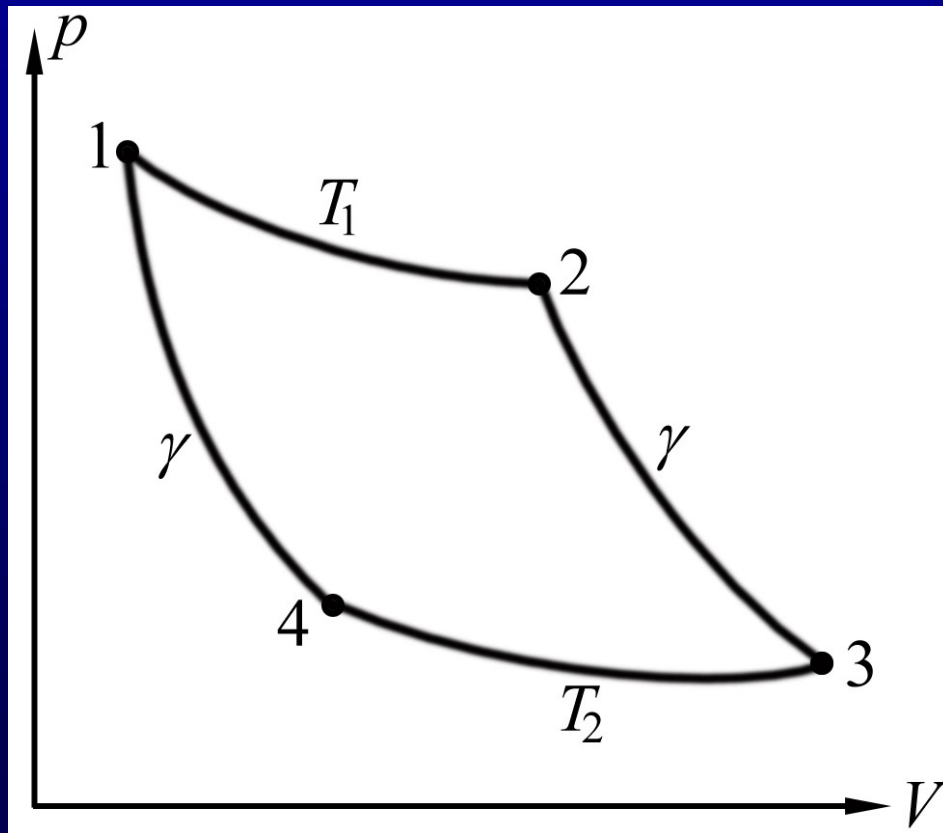
pistons

crankshaft

flywheel

11.6 Carnot cycle Carnot heat engine

$$W = \int p dV \quad \text{done by engine}$$



$$T_1, \quad U = U(T_1)$$

$$V_1 \rightarrow V_2 \quad \Delta W_1 > 0$$

$$\Delta Q_{12} \equiv Q_1 > 0$$

$$T_2, \quad \Delta Q_{34} < 0$$

$$-\Delta Q_{34} \equiv Q_2$$

For a complete cycle $\Delta U = 0$

$$\Delta Q_{12} + \Delta Q_{34} - \Delta W = 0$$

Efficiency is

$$\begin{aligned}\eta &= \frac{\Delta W}{\Delta Q_{12}} = \frac{\Delta Q_{12} + \Delta Q_{34}}{\Delta Q_{12}} \\ &= 1 + \frac{\Delta Q_{34}}{\Delta Q_{12}} \equiv 1 - \frac{Q_2}{Q_1}\end{aligned}$$

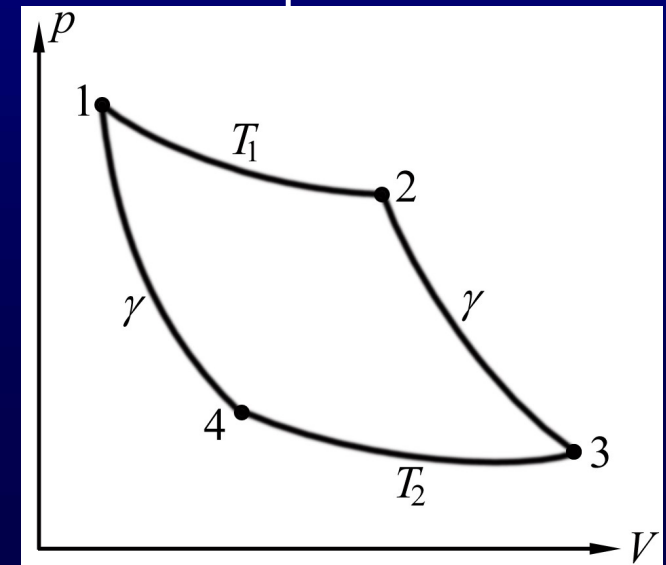
Along isotherm T_1 , $\Delta U = 0$

$$Q_1 \equiv \Delta Q_{12} = \Delta W_{12} = \int_{V_1}^{V_2} p dV = nRT_1 \ln \frac{V_2}{V_1}$$

Similarly

$$Q_2 = -\Delta Q_{34} = -\int_{V_3}^{V_4} p dV = nRT_2 \ln \frac{V_3}{V_4}$$

$$\eta = 1 - \frac{T_2 \ln \frac{V_3}{V_4}}{T_1 \ln \frac{V_2}{V_1}}$$



2 $\longrightarrow$ 3

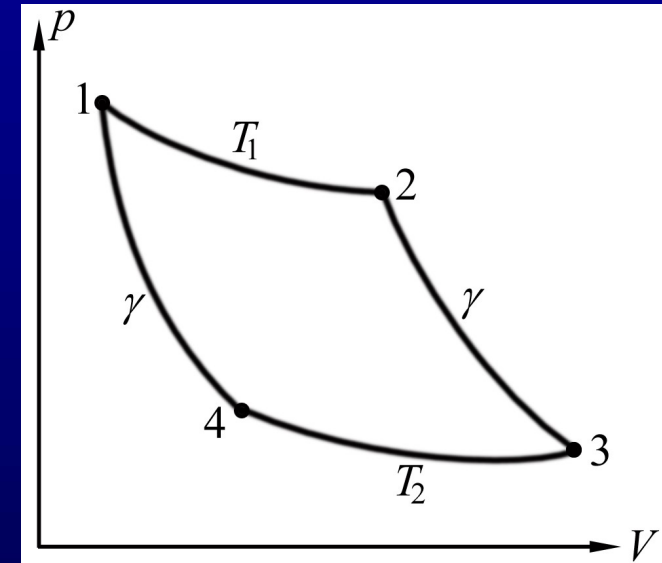
$$T_1 V_2^{\gamma-1} = T_2 V_3^{\gamma-1}$$

$$\frac{V_2}{V_1} = \frac{V_3}{V_4}$$

4 $\longrightarrow$ 1

$$T_1 V_1^{\gamma-1} = T_2 V_4^{\gamma-1}$$

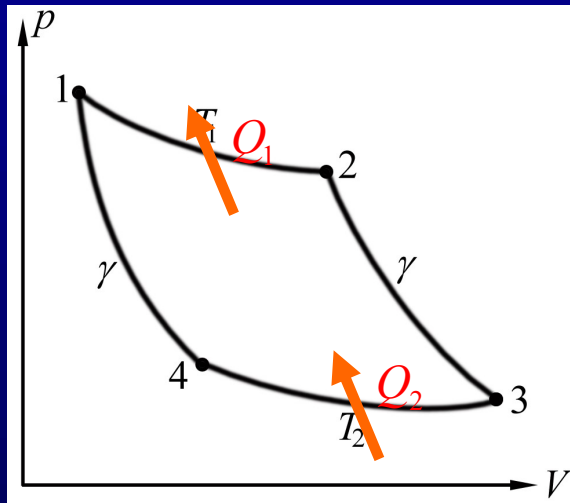
$$\eta = 1 - \frac{T_2}{T_1}$$



Assignment: 11.5, 11.9/11.10

Reverse Carnot Cycle

4-3-2-1-4



$$T_2, \quad V_4 \rightarrow V_3$$

$$Q_2 = \Delta Q_{43} = \int_{V_4}^{V_3} p dV = nRT_2 \ln \frac{V_3}{V_4}$$

$$T_1 \quad V_2 \rightarrow V_1$$

$$Q_1 = -\Delta Q_{21} = -\int_{V_2}^{V_1} p dV = nRT_1 \ln \frac{V_2}{V_1}$$

$$\eta_r = \frac{Q_2}{W} = \frac{Q_2}{Q_1 - Q_2} = \frac{T_2 \ln \frac{V_3}{V_4}}{T_1 \ln \frac{V_2}{V_1} - T_2 \ln \frac{V_3}{V_4}} = \frac{T_2}{T_1 - T_2}$$